

MATH 55 FINAL EXAM, PROF. SRIVASTAVA
DECEMBER 19, 2024, 11:40AM–2:30PM, 150 WHEELER.

Name: _____

SID: _____

GSI: _____

NAME OF THE STUDENT TO YOUR LEFT: _____

NAME OF THE STUDENT TO YOUR RIGHT: _____

INSTRUCTIONS: Write all answers clearly in the provided space. **Do not unstaple the exam.** Write proofs in complete English sentences. You can express your answers in terms of factorials and binomial coefficients without simplifying fractions. You may cite any theorem proved in class (but not HW and quizzes), by stating it or citing its name. **Fill in the multiple choice circles clearly.** You may use one 2–sided handwritten cheat sheet.

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Sign here: _____

Question	Points
1	15
2	10
3	10
4	10
5	10
6	10
7	9
8	10
9	16
Total:	100

Do not turn over this page until your instructor tells you to do so.
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1. Select true or false for each of the following by filling in a circle. There is no need to provide an explanation.

(a) (3 points) The compound proposition

$$(\forall x((p(x) \rightarrow q(x)) \wedge (\neg p(x) \rightarrow q(x))) \rightarrow \forall x q(x))$$

is a tautology.

Solution: True. Here is a proof: Assume the hypothesis. Let x be arbitrary. By universal instantiation, $((p(x) \rightarrow q(x)) \wedge (\neg p(x) \rightarrow q(x)))$. If $p(x)$ then we conclude that $q(x)$. Otherwise $\neg p(x)$, and we conclude $q(x)$. Thus, $q(x)$.

(b) (3 points) If A, B, C are nonempty sets, $f : A \rightarrow B$ is one-to-one, and $g : B \rightarrow C$ is one-to-one, then $g \circ f : A \rightarrow C$ must be one-to-one.

Solution: True. Assume $x, y \in A$ and $g \circ f(x) = g \circ f(y)$. This means $g(f(x)) = g(f(y))$. Since g is one-to-one, $f(x) = f(y)$. Since f is one-to-one, $x = y$, as desired.

(c) (3 points) For any three integers a, b, c , if $\gcd(a, b) = 1$ and $\gcd(b, c) = 1$ then $\gcd(a, c) = 1$.

Solution: False. For instance, take $a = 1$ and $b = c = 2$.

(d) (3 points) If a, b are nonzero integers then $(a + b)^5 \equiv a^5 + b^5 \pmod{5}$.

Solution: True. By the binomial theorem,

$$(a + b)^5 = a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5 \equiv a^5 + b^5 \pmod{5},$$

since $5 \equiv 10 \equiv 0 \pmod{5}$.

(e) (3 points) If X and Y are nonnegative random variables and $\mathbb{P}(X \leq Y) = 1$ then $V(X) \leq V(Y)$, where $V(\cdot)$ denotes the variance of a random variable.

Solution: False. For instance, let X be a Bernoulli(1/2) random variable and let $Y = 2$. Then $X \leq Y$ with probability 1 but $V(X) = 1/4, V(Y) = 0$.

[Scratch space below]

2. Assume a and b are irrational numbers with $a < b$.

- (a) (4 points) Prove that the function $f : (0, 1) \rightarrow (a, b)$ defined by $f(x) = a + (b - a)x$ is a bijection.

Solution: Suppose $y \in (a, b)$ is arbitrary. The equation $a + (b - a)x = y$ has exactly one solution, namely $x = (y - a)/(b - a)$. Thus, there is a unique $x \in (0, 1)$ such that $f(x) = y$, so f is both one-to-one and onto. *Alternatively, $g(y) = (y - a)/(b - a)$ is an inverse of f , so f is a bijection.*

- (b) (6 points) Prove that there exists an irrational number c such that $a < c < b$.

Solution: Assume for contradiction that no such c exists. Consider the set

$$S = \{c \in \mathbb{R} : a < c < b\} = (a, b).$$

By our assumption, $S \subseteq \mathbb{Q}$, the rational numbers. Since $\mathbb{Q}$ is countable, S is also countable (i.e., finite or countably infinite): we may see this by considering an enumeration of $\mathbb{Q}$ and crossing out every element not in S . By part (a), however, there is a bijection between S and $(0, 1)$, so $|S| = |(0, 1)|$. But by Cantor's theorem $(0, 1)$ is uncountable, implying that S is uncountable, a contradiction.

The above is a nonconstructive proof, for which part (a) was supposed to be a hint. There is also a constructive proof: consider $c = (a + b)/2$. If c is irrational we are done. Otherwise, c is rational. This implies that $(a + c)/2$ is irrational.

3. (a) (2 points) Prove that if x is an integer then $x^2 \equiv 0 \pmod{3}$ or $x^2 \equiv 1 \pmod{3}$.

Solution: Let x be an integer. Observe that x is congruent to 0, 1 or 2 (mod 3). If $x \equiv 0 \pmod{3}$ then $x^2 \equiv 0^2 \equiv 0 \pmod{3}$. If $x \equiv 1 \pmod{3}$ then $x^2 \equiv 1^2 \equiv 1 \pmod{3}$. If $x \equiv 2 \pmod{3}$ then $x^2 \equiv 4 \equiv 1 \pmod{3}$.

- (b) (3 points) Prove that if x and y are integers such that

$$x^2 + y^2 \equiv 3xy \pmod{3}$$

then $x \equiv y \equiv 0 \pmod{3}$. You may use the statement of part (a).

Solution:

Suppose $x^2 + y^2 \equiv 3xy \pmod{3}$. By part (a), each of x^2, y^2 is congruent to 0 or 1 (mod 3). If exactly one of $x^2 \equiv 1 \pmod{3}$ or $y^2 \equiv 1 \pmod{3}$ then $x^2 + y^2 \equiv 1 \not\equiv 0 \equiv 3(0)(0) \equiv 3xy \pmod{3}$. On the other hand, if both x and y are congruent to 1 (mod 3), then $x^2 + y^2 \equiv 2 \not\equiv 3(1)(1) \equiv 3xy \pmod{3}$. Thus, the only possibility is that $x \equiv y \equiv 0 \pmod{3}$, as desired.

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- (c) (5 points) Prove that the equation $x^2 + y^2 = 3xy$ has no *positive* integer solutions. (hint: proof by contradiction. Use the well-ordering principle and (b)).

Solution: Let

$$S = \{x + y : x, y \text{ are positive integers satisfying the given equation}\}.$$

Assume for contradiction that $S \neq \emptyset$. By the well-ordering principle, S has a least element and we can choose positive integers x, y such that $x + y$ is minimized and

$$x^2 + y^2 = 3xy.$$

By part (b) we can choose positive integers a, b such that $x = 3a, y = 3b$. Substituting, we have

$$9a^2 + 9b^2 = 3(3a)(3b) = 27ab.$$

Dividing both sides by 9 gives

$$a^2 + b^2 = 3ab,$$

implying that $a + b \in S$, contradicting the minimality of $x + y$.

4. For $n \geq 1$, let a_n denote the number of subsets of $\{1, \dots, n\}$ which do *not* contain two consecutive numbers (i.e., do not contain i and $i + 1$ for some i).
- (a) (2 points) What are a_1 and a_2 ?

Solution: Both $\emptyset, \{1\}$ do not contain consecutive elements, so $a_1 = 2$. For $n = 2$ the sets $\emptyset, \{1\}, \{2\}$ don't contain consecutive elements, so $a_2 = 3$.

- (b) (6 points) Derive a recurrence relation for $a_n, n \geq 3$.

Solution: Let $n \geq 3$ and let C_n be the set of subsets of $\{1, \dots, n\}$ which do not contain consecutive elements. An element $S \in C_n$ may be uniquely specified by the following process: (1) Decide whether $n \in S$ or $n \notin S$. (2) If $n \in S$, choose a subset $T \subseteq \{1, \dots, n - 1\}$ such that $n - 1 \notin T$ (to avoid consecutive elements $n - 1, n$ in S) and T does not contain consecutive elements, and let $S = T \cup \{n\}$. Note that this is the same as choosing $T \in C_{n-2}$. (3) If $n \notin S$, let S be an element of C_{n-1} .

By the sum rule, the total number of ways to execute this process is $|C_{n-2}| + |C_{n-1}|$. Thus, $a_n = a_{n-1} + a_{n-2}$.

Another way to do this is to notice that subsets of $\{1, \dots, n\}$ are in bijection with bit strings of length n , and the question is asking us to count strings which

do not contain two consecutive zeros, and example we did in class.

- (c) (2 points) Use your recurrence relation to compute the value of a_{10} .

Solution: Applying the recurrence, we obtain the sequence

$$a_1 = 2, 3, 5, 8, 13, 21, 34, 55, 89, 144 = a_{10}.$$

5. (a) (5 points) Prove that if G is a connected simple graph on n vertices with $n+1$ edges, then G must contain at least two distinct (i.e., not equal) cycles as subgraphs.

Solution: Assume G is a connected graph on n vertices. Choose a spanning tree T of G . Since T has n vertices it has exactly $n - 1$ edges by a theorem proven in class. Thus, there are exactly two edges of G which are not in T , call them $e_1 = xy$ and $e_2 = ab$. Observe that there is a unique path γ from x to y in T . The circuit $C_1 = \gamma, yx$ is now a cycle, as no vertices or edges are repeated. By the same argument, we obtain another cycle C_2 containing the edge e_2 and the unique path between its endpoints in T . These cycles are not the same because $e_1 \in C_1$ but $e_1 \notin C_2$. We conclude that G contains two cycles, as desired.

- (b) (5 points) Prove that if G is any simple graph on n vertices with $n + 1$ edges, then it must contain at least two distinct cycles. You may use the statement of part (a) even if you did not prove it.

Solution: Let $G_1, \dots, G_k$ be the connected components of G . Suppose they have $n_1, \dots, n_k$ vertices and $m_1, \dots, m_k$ edges, respectively. Observe that there must be some i for which $m_i \geq n_i + 1$; otherwise we would have $m_i \leq n_i$ for all i , yielding that the total number of edges of G is $m_1 + m_2 + \dots + m_k \leq n_1 + \dots + n_k = n$, which is not the case.

Choose a spanning tree T_i of G_i . Since $m_i \geq n_i + 1$ we can choose two edges e_1, e_2 of G_i not in T_i . By the same argument as in part (a), G_i contains two distinct cycles. This implies that G contains two distinct cycles, as desired.

6. Let n be an even positive integer.

- (a) (6 points) Prove that if G is a simple graph on n vertices in which every vertex has degree at least $n/2$, then G must be connected.

Solution: Let $G = (V, E)$ be a graph on n vertices with each vertex having degree at least $n/2$. Let $x, y \in V$ be arbitrary distinct vertices. We will show that there is a path in G between x and y .

If $xy \in E$ then there is a path of length 1 between x and y . Otherwise, let S be the set of vertices adjacent to x and T be the set of vertices adjacent to y in G , and notice that $|S| \geq n/2$, $|T| \geq n/2$ by our assumption. Observe that if $S \cap T = \emptyset$ then we would have $|S \cup T| = |S| + |T| = n$. However, $S \cup T \subseteq V - \{x, y\}$ so $|S \cup T| \leq (n-2)$, implying that this is not possible. Thus we can choose a vertex $z \in S \cap T$, i.e., a common neighbor of x and y . Now x, z, y is a path in G between x and y , as desired.

This is similar in spirit to the fact that every simple graph has two vertices which have the same degree, by the pigeonhole principle.

- (b) (4 points) Give an example of a disconnected simple graph G on n vertices in which every vertex has degree exactly $(n/2) - 1$.

Solution: Let G be a disjoint union of two complete graphs of size $n/2$. Then each vertex has degree $(n/2) - 1$, but the graph has two connected components so it is not connected. *In fact, this is the only example of a graph with the required properties. Note that we have actually described a family of graphs, one for every positive even n .*

7. Suppose Nikhil draws a hand of 5 cards uniformly at random from a deck of 52 (13 cards $A, 2, \dots, 10, J, Q, K$ of each of the four suits: hearts, spades, diamonds, clubs).

- (a) (3 points) What is the probability that the hand is a flush, that is, five cards of the same suit?

Solution: Let the sample space S be the set of 5-combinations of chosen from 52 cards. A flush is uniquely specified by choosing (1) a suit (4 possibilities) (2) an unordered hand of 5 cards from that suit. By the product rule the total number of flushes is $4 \binom{13}{5}$. The probability of a flush is therefore $\frac{4 \binom{13}{5}}{\binom{52}{5}}$.

- (b) (3 points) What is the probability that the hand contains at least one ace?

Solution: Let E be the event that the hand contains at least one ace. We will calculate $\mathbb{P}(\overline{E})$. A hand containing no ace is uniquely specified by choosing 5 cards from the 48 non-ace cards in the deck, so $|\overline{E}| = \binom{48}{5}$. Consequently,

$$\mathbb{P}(E) = 1 - \frac{\binom{48}{5}}{\binom{52}{5}}.$$

- (c) (3 points) What is the probability that the hand is a flush *and* contains at least one ace?

Solution: A flush which contains an ace is specified as follows: (1) Choose a suit (2) Choose 4 non-ace cards from that suit. (3) Choose an ace from that suit. By the product rule, the number of ways to specify such a hand is $4\binom{12}{4} \cdot 1$. Thus, the desired probability is $4\binom{12}{4}/\binom{52}{5}$.

8. Consider the following experiment: Oski rolls a fair six-sided die and observes a uniformly random outcome $D \in \{1, 2, \dots, 6\}$. After that, Oski flips D independent fair (i.e., equally likely to be heads or tails) coins. Let C be the number of heads observed.

- (a) (3 points) What is $\mathbb{P}(C = 5|D = 6)$?

Solution: Conditioned on $\{D = 6\}$, C has a *Binomial*(6, 1/2) distribution. thus,

$$\mathbb{P}(C = 5|D = 6) = \binom{6}{5} \frac{1}{2^6} = \frac{6}{64} = \frac{3}{32}.$$

- (b) (4 points) What is $\mathbb{P}(D = 6|C = 5)$?

Solution: We will apply Bayes' rule. To do that we first calculate $\mathbb{P}(D = 6) = 1/6$ and

$$\mathbb{P}(C = 5) = \sum_{i=1}^6 \mathbb{P}(C = 5|D = i)\mathbb{P}(D = i)$$

where we used the law of total probability. Notice that $\mathbb{P}(C = 5|D = i) = 0$ for $i = 1, 2, 3, 4$ since it is impossible to observe 5 heads when fewer than 5 coins are flipped. Moreover, $\mathbb{P}(D = i) = 1/6$ for every i . Combining these, we obtain

$$\mathbb{P}(C = 5) = \mathbb{P}(C = 5|D = 5)(1/6) + \mathbb{P}(C = 5|D = 6)(1/6).$$

Conditioned on $\{D = 5\}$, C has a *Binomial*(5, 1/2) distribution so $\mathbb{P}(C = 5|D = 5) = \binom{5}{5} \frac{1}{2^5} = 1/32$. We already calculated $\mathbb{P}(C = 5|D = 6)$ in part (a). Plugging these in, we have

$$\mathbb{P}(C = 5) = (1/6)(1/32 + 3/32) = \frac{4}{6 \cdot 32} = \frac{1}{48}.$$

By Bayes' rule we now have

$$\mathbb{P}(D = 6|C = 5) = \frac{\mathbb{P}(C = 5|D = 6)\mathbb{P}(D = 6)}{\mathbb{P}(C = 5)} = (3/32) \frac{1/6}{1/48} = \frac{24}{32} = \frac{3}{4}.$$

- (c) (3 points) For what value of x is $\mathbb{P}(D = x|C = 5)$ maximized?

Solution: First, we have

$$\sum_{x=1}^6 \mathbb{P}(D = x|C = 5) = 1,$$

by considering the experiment conditioned on $\{C = 5\}$. Notice that if $x = 1, 2, 3, 4$ then $\mathbb{P}(D = x|C = 5) = \mathbb{P}(C = 5|D = x) \frac{\mathbb{P}(D=x)}{\mathbb{P}(C=5)} = 0$ since $\mathbb{P}(C = 5|D = x) = 0$ for these values of x . So the only nonzero values are $x = 5, 6$. Thus, $\mathbb{P}(D = 5|C = 5) + \mathbb{P}(D = 6|C = 5) = 1$. By part (b) we have $\mathbb{P}(D = 5|C = 5) = 1 - (3/4) = 1/4$, so the maximizing value is $x = 6$.

9. Assume that there are 500 lockers labeled $\{1, \dots, 500\}$ at the Recreational Sports Facility (RSF) at Cal (which I highly recommend, if you haven't been there). Assume each one has a combination lock whose code is an independent uniformly random 4-digit string with digits in $\{0, 1, \dots, 9\}$. Call an unordered pair of lockers *vulnerable* if their locks have the same code.

- (a) (3 points) For $i, j \in \{1, \dots, 500\}$ with $i < j$, let E_{ij} denote the event that the pair $\{i, j\}$ is vulnerable. Calculate $\mathbb{P}(E_{12})$.

Solution: The code for each locker is chosen uniformly and independently from 10^4 choices. Let $X_i, i = 1, \dots, 500$ be random variables indicating the codes received by Lockers $i = 1, \dots, 500$. By the law of total probability,

$$\mathbb{P}(E_{12}) = \sum_{i=1}^{10^4} \mathbb{P}(E_{12}|X_1 = i) \cdot (1/10^4) = \sum_{i=1}^{10^4} \mathbb{P}(X_2 = i|X_1 = i) = \sum_{i=1}^{10^4} (1/10^4)^2 = 1/10^4,$$

where we used that X_1 and X_2 are independent.

- (b) (2 points) Are the events E_{12} and E_{34} independent? Justify your answer.

Solution: We calculate

$$\mathbb{P}(E_{12}|E_{34}) = \frac{\mathbb{P}(E_{12} \cap E_{34})}{\mathbb{P}(E_{34})} = \frac{|E_{12} \cap E_{34}|}{|E_{34}|} = (1/10^4)^2 / (1/10^4),$$

where we used that $|E_{12} \cap E_{34}| = (10^4)^{500-2}$ since such a locker assignment is uniquely specified by $X_1, X_3, X_5, X_6, \dots, X_{500}$. On the other hand, by part (a) we have $\mathbb{P}(E_{12}) = 1/10^4 = \mathbb{P}(E_{12}|E_{34})$. Thus, these events are independent.

This should also be conceptually clear, since the combinations on Lockers 1 and 2 are independent of those on 3 and 4.

- (c) (2 points) Are the events E_{12} and E_{23} independent? Justify your answer.

Solution: Again, we calculate:

$$\mathbb{P}(E_{12}|E_{23}) = \frac{\mathbb{P}(E_{12} \cap E_{23})}{\mathbb{P}(E_{23})}.$$

An outcome is in $E_{12} \cap E_{23}$ iff $X_1 = X_2 = X_3$, so such an outcome is uniquely specified by $X_1, X_4, X_5, \dots, X_{500}$, from which we conclude $|E_{12} \cap E_{23}| = (10^4)^{500-2}$. Thus,

$$\mathbb{P}(E_{12}|E_{23}) = \frac{\mathbb{P}(E_{12} \cap E_{23})}{\mathbb{P}(E_{23})} = 1/(10^4) = \mathbb{P}(E_{12}),$$

so the events are independent.

- (d) (4 points) Let X be a random variable equal to the number of vulnerable pairs. Calculate the expected value $\mathbb{E}X$.

Solution: Let P denote the set of unordered pairs of lockers. Note that $|P| = \binom{500}{2}$. For each $ij \in P$ (shorthand for $\{i, j\}$), define X_{ij} to be the indicator random variable of E_{ij} . Then, $X = \sum_{ij \in P} X_{ij}$. Note that

$$\mathbb{E}X_{ij} = \mathbb{P}(E_{ij}) = 1/(10^4)$$

as calculated in part (a). By linearity of expectation,

$$\mathbb{E}X = \sum_{ij \in P} \mathbb{E}X_{ij} = \binom{500}{2} \frac{1}{10^4}.$$

- (e) (5 points) Calculate the variance $V(X)$.

Solution: By part (d) we have

$$X = \sum_{ij \in P} X_{ij}.$$

We showed in part (c) that for distinct pairs $ij \in P, i'j' \in P$ the events $E_{ij}, E_{i'j'}$ are independent. This implies that the random variables X_{ij} for $ij \in P$ are pairwise independent. In Lecture 26 we showed that the variance of a sum of pairwise independent random variables is the sum of the variances. Therefore,

$$V(X) = \sum_{ij \in P} V(X_{ij}) = \binom{500}{2} V(X_{12}),$$

since the random variables X_{ij} all have the same distribution as X_{12} . Since X_{12}

is an indicator random variable, we have

$$\mathbb{E}X_{12}^2 = \mathbb{E}X_{12},$$

and

$$V(X_{12}) = \mathbb{E}X_{12} - (\mathbb{E}X_{12})^2 = (1/10^4)(1 - 1/10^4).$$

We conclude that

$$V(X) = \binom{500}{2} (1/10^4)(1 - 1/10^4).$$

This was the hardest question on the exam, because you had to realize that E_{12} and E_{23} are pairwise independent (which is a bit surprising), use this to conclude that the corresponding random variables are pairwise independent, and know that variances add up for pairwise independent random variables.

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[Scratch Paper 3]